

Answers to Question 1:

- a) My name is Ashish Pant. People call me by my first name - Ashish.
- b) My major is Applied Statistics. I started the MAS program in the Spring 26
- c) I have completed just one course - STAT-5279 during the Spring26 semester. In general, any form of mathematics is difficult for me, including Statistics. This is why, I am taking one course at a time for the time being.
- d) I am a Quantitative Trader since 2014. I have extensive data engineering experience. My market data repository is ~31 TB. I have tried to use statistics to study market data. I am familiar with descriptive statistics, plotting, types of plots, binning etc. Most of my statistics experience is related to EDA but most of the time, I had no clue what I was doing which is primarily why I am in this program. STAT-5279 was an eye opener.
- e) After STAT-5279, I am now using R and Positron comfortably. I am comfortable with ggplot and Quarto. I am a decent Python, Java, C++ and C# programmer.

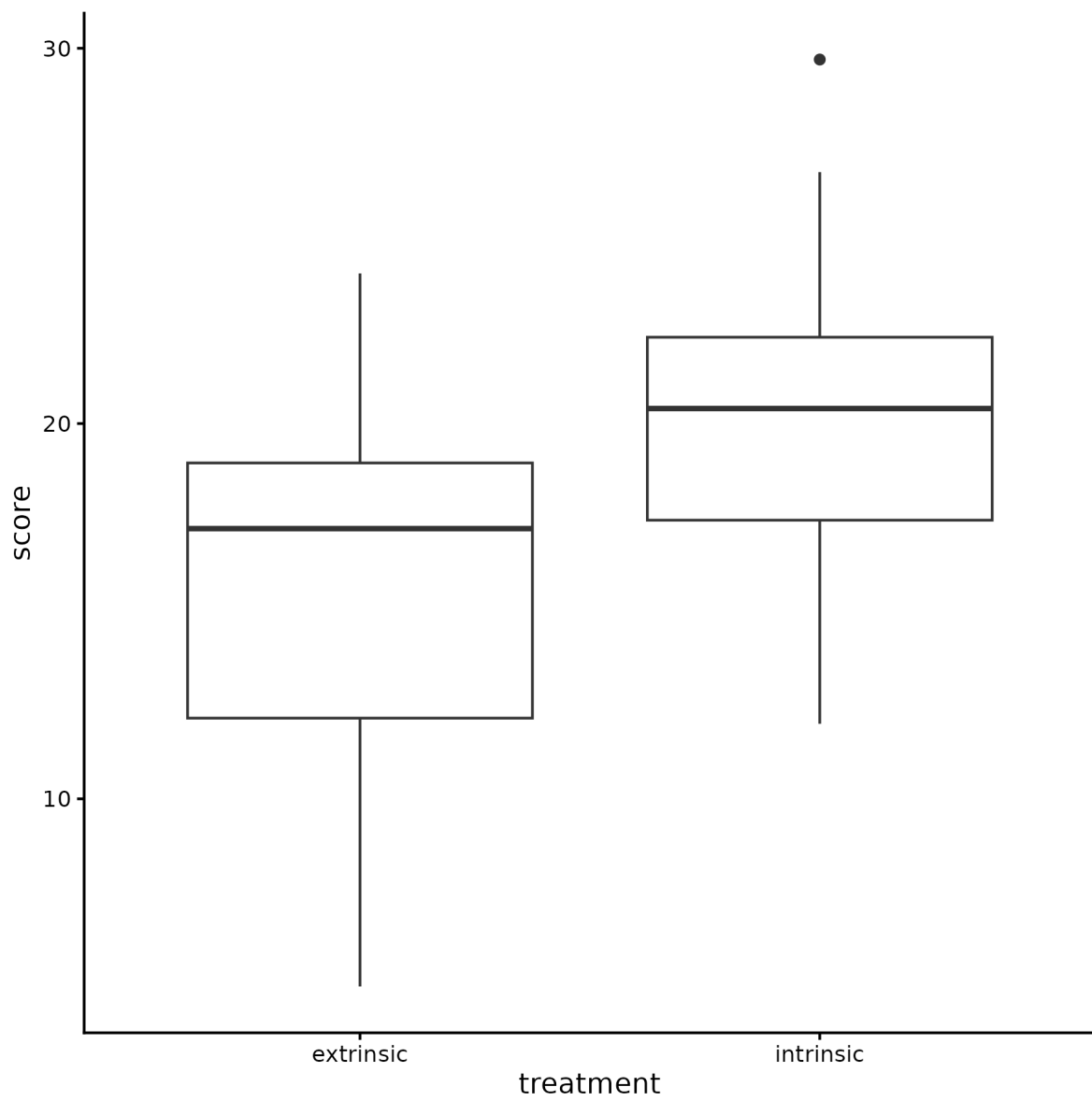
Answers to Question 2:

- a) In 2023, I developed models to trade commodities futures markets based exclusively on binning. For each variable in my historical dataset. I used Python packages like optbinning to best describe the relationship between that variable and market price. Then, I created ensembles of scikit-learn models and tested against hold-out (unseen) data. I generated backtests based on hypothetical results to confirm commercial feasibility. And then, I made the worst decision of my career - I decided to use these models to actually trade the markets. As you can probably guess, these models failed to perform when subjected to live data.
- b) Adventure.
- c) Uber/Lyft driver, Grocery Shopper.

Answers to Question 3

- a) If permitted, I would like to work independently on the project. I would like to use a custom dataset from my professional background in futures trading.
- b) I have extensive experience working with large amounts of data. I learnt R in STAT-5279 recently. I have been programming for 30+ years in Python, Java, C++.

Answer to Question 4:



Box-plot showing scores for each treatment